

[see Fig. 10.33(e)], but their separation is enough so that the depth of the valley between them is sufficient to make the modes easy to separate. A threshold placed midway between the two peaks would do the job. Figure 10.33(c) shows the result of corrupting the image with Gaussian noise of zero mean and a standard deviation of 50 intensity levels. As the histogram in Fig. 10.33(f) shows, the situation is much more serious now, as there is no way to differentiate between the two modes. Without additional processing (such as the methods discussed later in this section) we have little hope of finding a suitable threshold for segmenting this image.

The Role of Illumination and Reflectance in Image Thresholding

Figure 10.34 illustrates the effect that illumination can have on the histogram of an image. Figure 10.34(a) is the noisy image from Fig. 10.33(b), and Fig. 10.34(d) shows its histogram. As before, this image is easily segmentable with a single threshold. With reference to the image formation model discussed in Section 2.3, suppose that we multiply the image in Fig. 10.34(a) by a nonuniform intensity function, such as the intensity ramp in Fig. 10.37(b), whose histogram is shown in Fig. 10.34(e). Figure 10.34(c) shows the product of these two images, and Fig. 10.34(f) is the resulting histogram. The deep valley between peaks was corrupted to the point where separation of the modes without additional processing (to be discussed later in this section) is no longer possible. Similar results would be obtained if the illumination was

In theory, the histogram of a ramp image is uniform. In practice, the degree of uniformity depends on the size of the image and number of intensity levels.

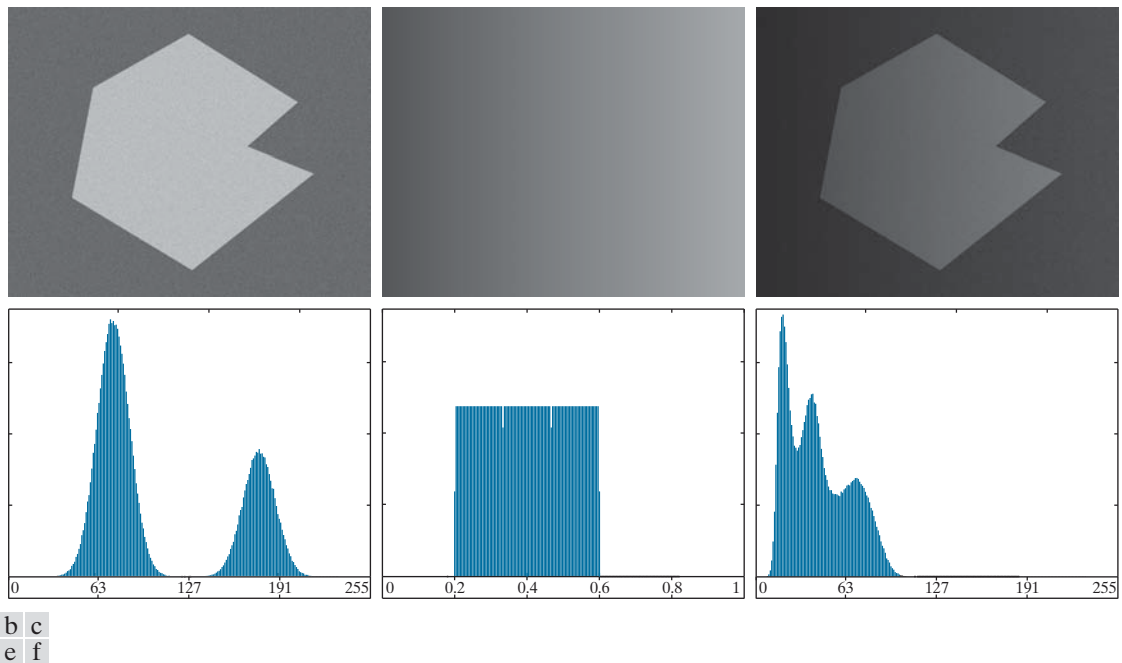


FIGURE 10.34 (a) Noisy image. (b) Intensity ramp in the range $[0.2, 0.6]$. (c) Product of (a) and (b). (d) through (f) Corresponding histograms.

perfectly uniform, but the reflectance of the image was not, as a result, for example, of natural reflectivity variations in the surface of objects and/or background.

The important point is that illumination and reflectance play a central role in the success of image segmentation using thresholding or other segmentation techniques. Therefore, controlling these factors when possible should be the first step considered in the solution of a segmentation problem. There are three basic approaches to the problem when control over these factors is not possible. The first is to correct the shading pattern directly. For example, nonuniform (but fixed) illumination can be corrected by multiplying the image by the inverse of the pattern, which can be obtained by imaging a flat surface of constant intensity. The second is to attempt to correct the global shading pattern via processing using, for example, the top-hat transformation introduced in Section 9.8. The third approach is to “work around” nonuniformities using variable thresholding, as discussed later in this section.

BASIC GLOBAL THRESHOLDING

When the intensity distributions of objects and background pixels are sufficiently distinct, it is possible to use a single (*global*) threshold applicable over the entire image. In most applications, there is usually enough variability between images that, even if global thresholding is a suitable approach, an algorithm capable of estimating the threshold value for each image is required. The following iterative algorithm can be used for this purpose:

1. Select an initial estimate for the global threshold, T .
2. Segment the image using T in Eq. (10-46). This will produce two groups of pixels: G_1 , consisting of pixels with intensity values $> T$; and G_2 , consisting of pixels with values $\leq T$.
3. Compute the average (mean) intensity values m_1 and m_2 for the pixels in G_1 and G_2 , respectively.
4. Compute a new threshold value midway between m_1 and m_2 :

$$T = \frac{1}{2}(m_1 + m_2)$$

5. Repeat Steps 2 through 4 until the difference between values of T in successive iterations is smaller than a predefined value, ΔT .

The algorithm is stated here in terms of successively thresholding the input image and calculating the means at each step, because it is more intuitive to introduce it in this manner. However, it is possible to develop an equivalent (and more efficient) procedure by expressing all computations in the terms of the image histogram, which has to be computed only once (see Problem 10.29).

The preceding algorithm works well in situations where there is a reasonably clear valley between the modes of the histogram related to objects and background. Parameter ΔT is used to stop iterating when the changes in threshold values is small. The initial threshold must be chosen greater than the minimum and less than the maximum intensity level in the image (the average intensity of the image is a good

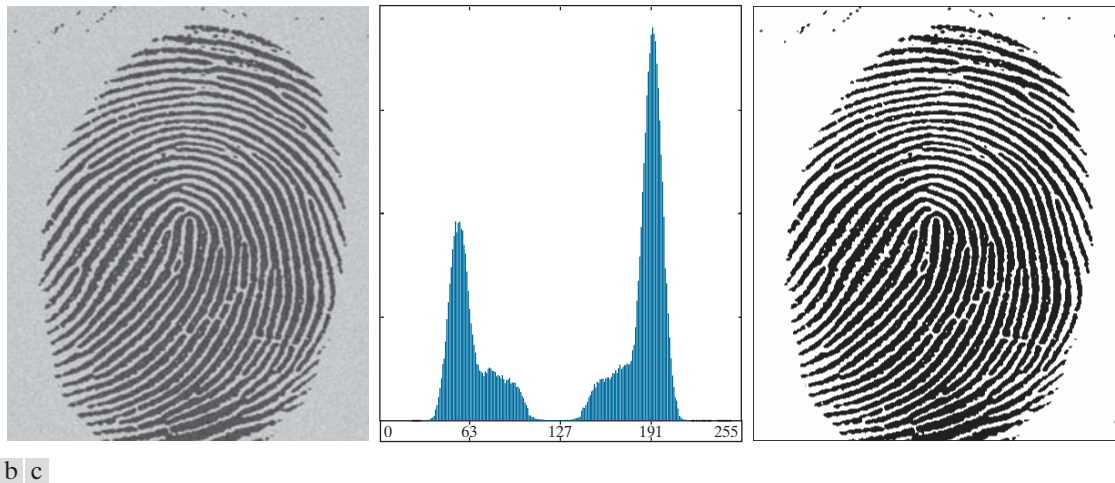


FIGURE 10.35 (a) Noisy fingerprint. (b) Histogram. (c) Segmented result using a global threshold (thin image border added for clarity). (Original image courtesy of the National Institute of Standards and Technology.).

initial choice for T). If this condition is met, the algorithm converges in a finite number of steps, whether or not the modes are separable (see Problem 10.30).

EXAMPLE 10.13: Global thresholding.

Figure 10.35 shows an example of segmentation using the preceding iterative algorithm. Figure 10.35(a) is the original image and Fig. 10.35(b) is the image histogram, showing a distinct valley. Application of the basic global algorithm resulted in the threshold $T = 125.4$ after three iterations, starting with T equal to the average intensity of the image, and using $\Delta T = 0$. Figure 10.35(c) shows the result obtained using $T = 125$ to segment the original image. As expected from the clear separation of modes in the histogram, the segmentation between object and background was perfect.

OPTIMUM GLOBAL THRESHOLDING USING OTSU'S METHOD

Thresholding may be viewed as a statistical-decision theory problem whose objective is to minimize the average error incurred in assigning pixels to two or more groups (also called *classes*). This problem is known to have an elegant closed-form solution known as the *Bayes decision function* (see Section 12.4). The solution is based on only two parameters: the probability density function (PDF) of the intensity levels of each class, and the probability that each class occurs in a given application. Unfortunately, estimating PDFs is not a trivial matter, so the problem usually is simplified by making workable assumptions about the form of the PDFs, such as assuming that they are Gaussian functions. Even with simplifications, the process of implementing solutions using these assumptions can be complex and not always well-suited for real-time applications.

The approach in the following discussion, called *Otsu's method* (Otsu [1979]), is an attractive alternative. The method is optimum in the sense that it maximizes the

between-class variance, a well-known measure used in statistical discriminant analysis. The basic idea is that properly thresholded classes should be distinct with respect to the intensity values of their pixels and, conversely, that a threshold giving the best separation between classes in terms of their intensity values would be the best (optimum) threshold. In addition to its optimality, Otsu's method has the important property that it is based entirely on computations performed on the histogram of an image, an easily obtainable 1-D array (see Section 3.3).

Let $\{0, 1, 2, \dots, L-1\}$ denote the set of L distinct integer intensity levels in a digital image of size $M \times N$ pixels, and let n_i denote the number of pixels with intensity i . The total number, MN , of pixels in the image is $MN = n_0 + n_1 + n_2 + \dots + n_{L-1}$. The normalized histogram (see Section 3.3) has components $p_i = n_i/MN$, from which it follows that

$$\sum_{i=0}^{L-1} p_i = 1 \quad p_i \geq 0 \quad (10-48)$$

Now, suppose that we select a threshold $T(k) = k$, $0 < k < L-1$, and use it to threshold the input image into two classes, c_1 and c_2 , where c_1 consists of all the pixels in the image with intensity values in the range $[0, k]$ and c_2 consists of the pixels with values in the range $[k+1, L-1]$. Using this threshold, the probability, $P_1(k)$, that a pixel is assigned to (i.e., thresholded into) class c_1 is given by the cumulative sum

$$P_1(k) = \sum_{i=0}^k p_i \quad (10-49)$$

Viewed another way, this is the probability of class c_1 occurring. For example, if we set $k = 0$, the probability of class c_1 having any pixels assigned to it is zero. Similarly, the probability of class c_2 occurring is

$$P_2(k) = \sum_{i=k+1}^{L-1} p_i = 1 - P_1(k) \quad (10-50)$$

From Eq. (3-25), the *mean intensity* value of the pixels in c_1 is

$$\begin{aligned} m_1(k) &= \sum_{i=0}^k iP(i/c_1) = \sum_{i=0}^k iP(c_1/i)P(i)/P(c_1) \\ &= \frac{1}{P_1(k)} \sum_{i=0}^k ip_i \end{aligned} \quad (10-51)$$

where $P_1(k)$ is given by Eq. (10-49). The term $P(i/c_1)$ in Eq. (10-51) is the probability of intensity value i , given that i comes from class c_1 . The rightmost term in the first line of the equation follows from Bayes' formula:

$$P(A/B) = P(B/A)P(A)/P(B)$$

The second line follows from the fact that $P(c_1/i)$, the probability of c_1 given i , is 1 because we are dealing only with values of i from class c_1 . Also, $P(i)$ is the probability of the i th value, which is the i th component of the histogram, p_i . Finally, $P(c_1)$ is the probability of class c_1 which, from Eq. (10-49), is equal to $P_1(k)$.

Similarly, the *mean intensity* value of the pixels assigned to class c_2 is

$$\begin{aligned} m_2(k) &= \sum_{i=k+1}^{L-1} iP(i/c_2) \\ &= \frac{1}{P_2(k)} \sum_{i=k+1}^{L-1} ip_i \end{aligned} \quad (10-52)$$

The *cumulative mean* (average intensity) up to level k is given by

$$m(k) = \sum_{i=0}^k ip_i \quad (10-53)$$

and the average intensity of the entire image (i.e., the *global mean*) is given by

$$m_G = \sum_{i=0}^{L-1} ip_i \quad (10-54)$$

The validity of the following two equations can be verified by direct substitution of the preceding results:

$$P_1m_1 + P_2m_2 = m_G \quad (10-55)$$

and

$$P_1 + P_2 = 1 \quad (10-56)$$

where we have omitted the ks temporarily in favor of notational clarity.

In order to evaluate the effectiveness of the threshold at level k , we use the normalized, dimensionless measure

$$\eta = \frac{\sigma_B^2}{\sigma_G^2} \quad (10-57)$$

where σ_G^2 is the *global variance* [i.e., the intensity variance of all the pixels in the image, as given in Eq. (3-26)],

$$\sigma_G^2 = \sum_{i=0}^{L-1} (i - m_G)^2 p_i \quad (10-58)$$

and σ_B^2 is the *between-class variance*, defined as

$$\sigma_B^2 = P_1(m_1 - m_G)^2 + P_2(m_2 - m_G)^2 \quad (10-59)$$

This expression can also be written as

$$\begin{aligned} \sigma_B^2 &= P_1P_2(m_1 - m_2)^2 \\ &= \frac{(m_GP_1 - m)^2}{P_1(1 - P_1)} \end{aligned} \quad (10-60)$$

The second step in this equation makes sense only if P_1 is greater than 0 and less than 1, which, in view of Eq. (10-56), implies that P_2 must satisfy the same condition.